

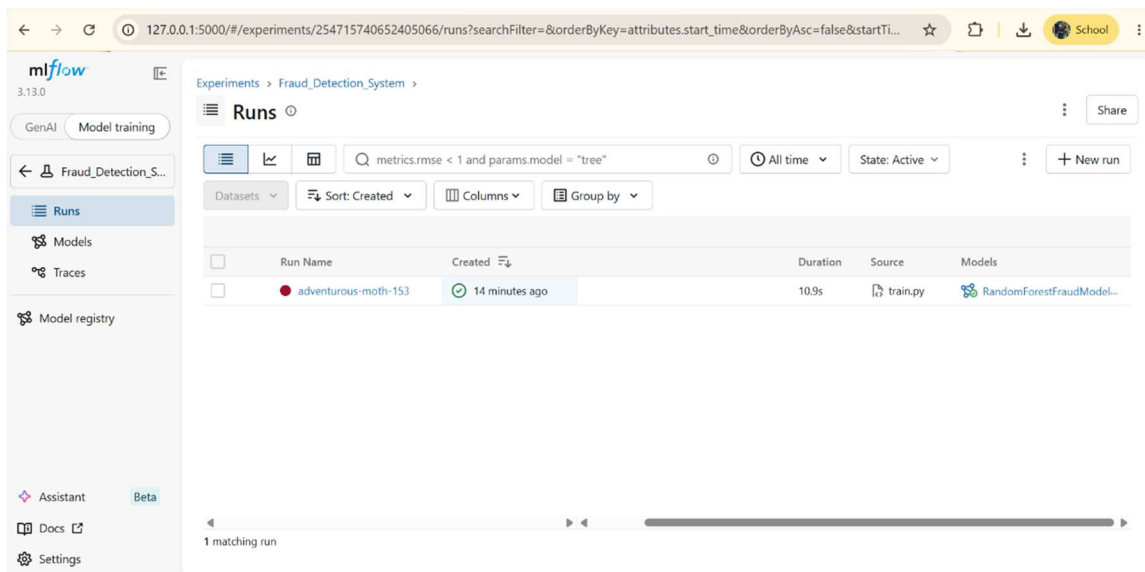
THE UNIVERSITY OF LAHORE

Department of Intelligent Systems - Spring Semester 2026

Name:	Ahad Ali	Subject:	MLOPS
Section:	BSES	Project # :	4
Sap ID:	70150510	Submitted To:	Sir Sikander Hayat

Model Training & Experiment Tracking (MLflow) :

MLflow Experiment Dashboard & Run Logs:



The initial phase of the pipeline focuses on experiment tracking and model lineage using MLflow. As shown in the dashboard, a training run named 'adventurous-moth-153' was logged successfully via train.py.

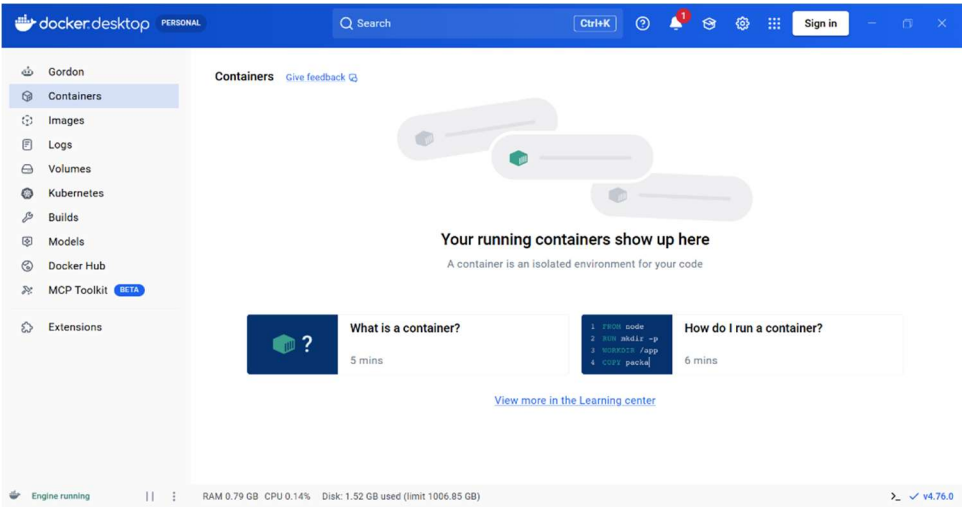
Production Workspace & Environment Setup :

Clean Production Directory Structure:

Name	Date modified	Type	Size
tracking_data	6/4/2026 12:19 AM	File folder	
deploy	6/4/2026 4:39 PM	Windows PowerShell ...	2 KB
Dockerfile	6/4/2026 4:30 PM	File	1 KB
main.py	6/4/2026 12:54 AM	Python.File	3 KB
model.pkl	6/4/2026 12:19 AM	PKL File	382 KB
requirements	6/4/2026 12:58 AM	Text Document	1 KB
train.py	6/4/2026 12:18 AM	Python.File	2 KB

The final directory containing only the localized tracking data, execution scripts (main.py, train.py), environment (requirements.txt), the model binary (model.pkl), and configuration files (Dockerfile, deploy.ps1).

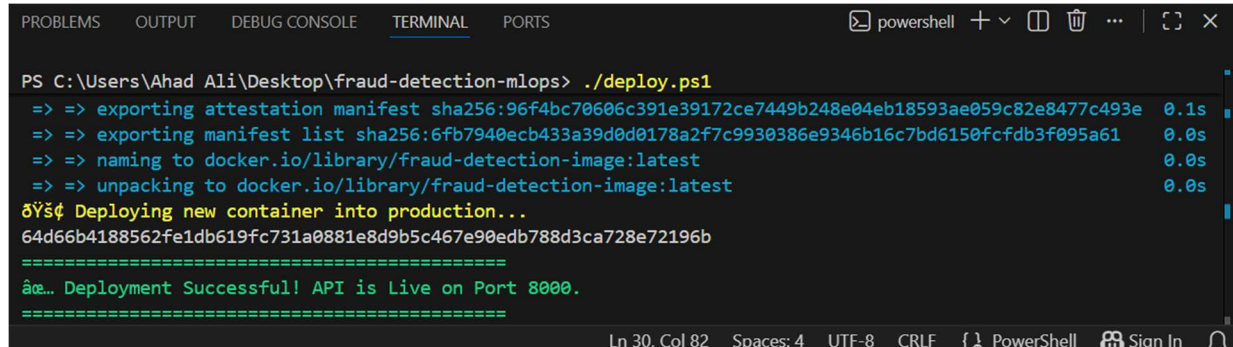
Containerization Environment (Docker Engine Status):



To achieve platform independence and isolated execution, Docker Desktop was initialized locally.

Automated CI/CD Deployment :

PowerShell Automated Pipeline Execution:

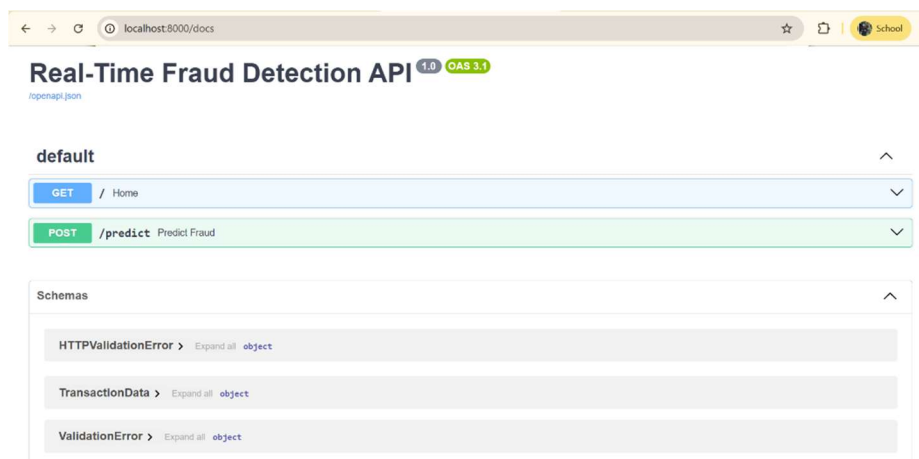


```
PS C:\Users\Ahad Ali\Desktop\fraud-detection-mlops> ./deploy.ps1
=> => exporting attestation manifest sha256:96f4bc70606c391e39172ce7449b248e04eb18593ae059c82e8477c493e 0.1s
=> => exporting manifest list sha256:6fb7940ecb433a39d0d0178a2f7c9930386e9346b16c7bd6150fcfdb3f095a61 0.0s
=> => naming to docker.io/library/fraud-detection-image:latest 0.0s
=> => unpacking to docker.io/library/fraud-detection-image:latest 0.0s
ðŸš€ Deploying new container into production...
64d66b4188562fe1db619fc731a0881e8d9b5c467e90edb788d3ca728e72196b
=====
âœ… Deployment Successful! API is Live on Port 8000.
=====
```

A custom CI/CD automation script (deploy.ps1) was developed to manage the deployment lifecycle. The console logs confirm a flawless termination with the confirmation status: "Deployment Successful! API is Live on Port 8000".

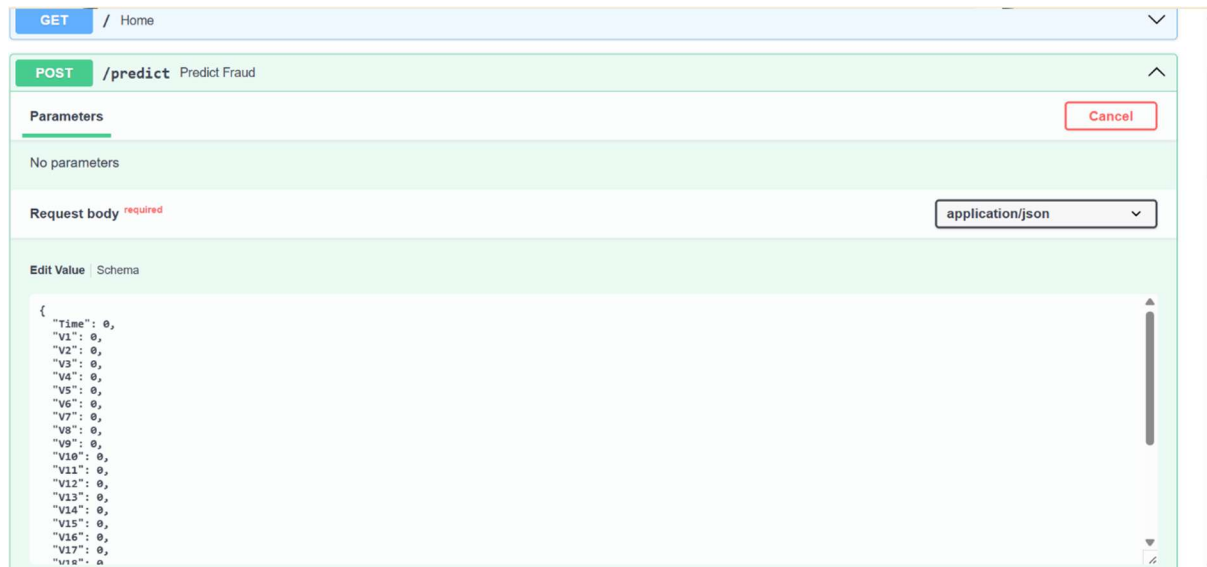
Production API Serving & Validation (FastAPI) :

FastAPI Swagger UI Interface:



With the container running seamlessly, the REST API endpoints are exposed on port 8000. The UI displays the exposed route endpoints (GET/HOME and POST/PREDICT) mapped to validate payload schemas dynamically.

Endpoint Schema & Request Payload Configuration:



GET / Home

POST /predict Predict Fraud

Parameters

No parameters

Request body **required** application/json

Edit Value Schema

```
{
  "Time": 0,
  "V1": 0,
  "V2": 0,
  "V3": 0,
  "V4": 0,
  "V5": 0,
  "V6": 0,
  "V7": 0,
  "V8": 0,
  "V9": 0,
  "V10": 0,
  "V11": 0,
  "V12": 0,
  "V13": 0,
  "V14": 0,
  "V15": 0,
  "V16": 0,
  "V17": 0,
  "V18": 0
}
```

Expanding the (Post / Predict) route reveals the structural JSON schema expected by the machine learning backend.

Automated Request Compilation (cURL Command):



Responses

Curl

```
curl -X 'POST' \
'http://127.0.0.1:8000/predict' \
-H 'accept: application/json' \
-H 'Content-type: application/json' \
-d '{
  "Time": 0,
  "V1": 0,
  "V2": 0,
  "V3": 0,
  "V4": 0,
  "V5": 0,
  "V6": 0,
  "V7": 0,
  "V8": 0,
  "V9": 0,
  "V10": 0,
  "V11": 0,
  "V12": 0,
  "V13": 0,
  "V14": 0,
  "V15": 0,
  "V16": 0,
  "V17": 0,
  "V18": 0
}'
```

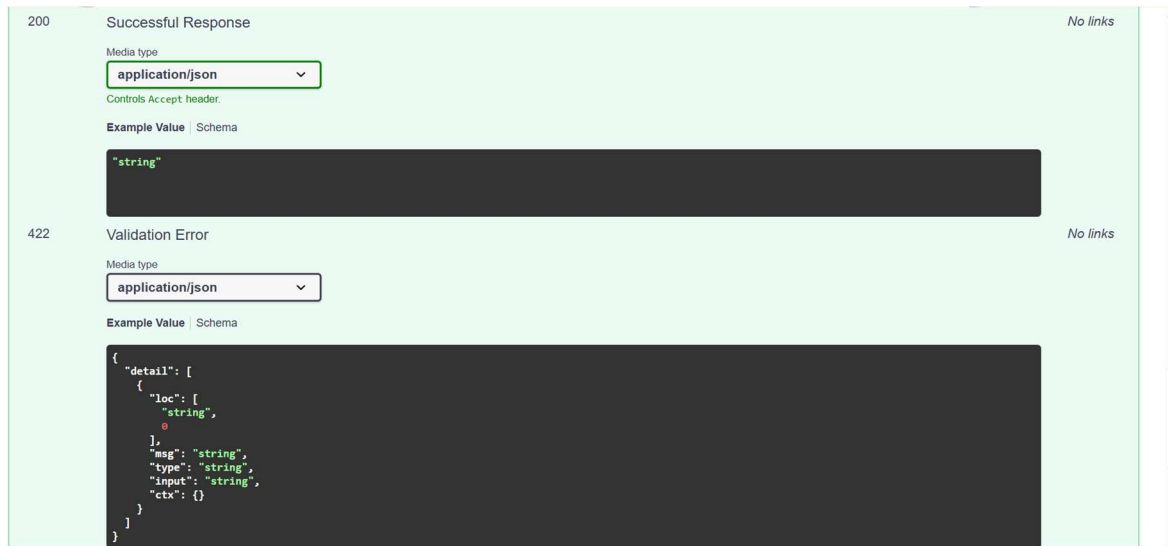
Request URL

http://127.0.0.1:8000/predict

Server response

The API engine auto-compiles the transaction input payload parameters into an executive cURL request mapped directly against the loopback address destination ([http://127.0.0.1:8000/predict] (<http://127.0.0.1:8000/predict>)).

Model Testing and Real-Time Fraud Prediction Verification



To ensure proper routing, model inference responses are thoroughly checked. The system evaluates incoming transactions against HTTP validation error criteria (422 Unprocessable Entity logs)

Live Output Inference Result (200 OK Response):



The framework successfully returned an HTTP 200 OK status code response in just a few milliseconds. The JSON response object contains the model's live decision output payload: ("prediction_code" : 0, "status" : "Legitimate"). This confirms the entire pipeline from training to automated container serving is fully functional.

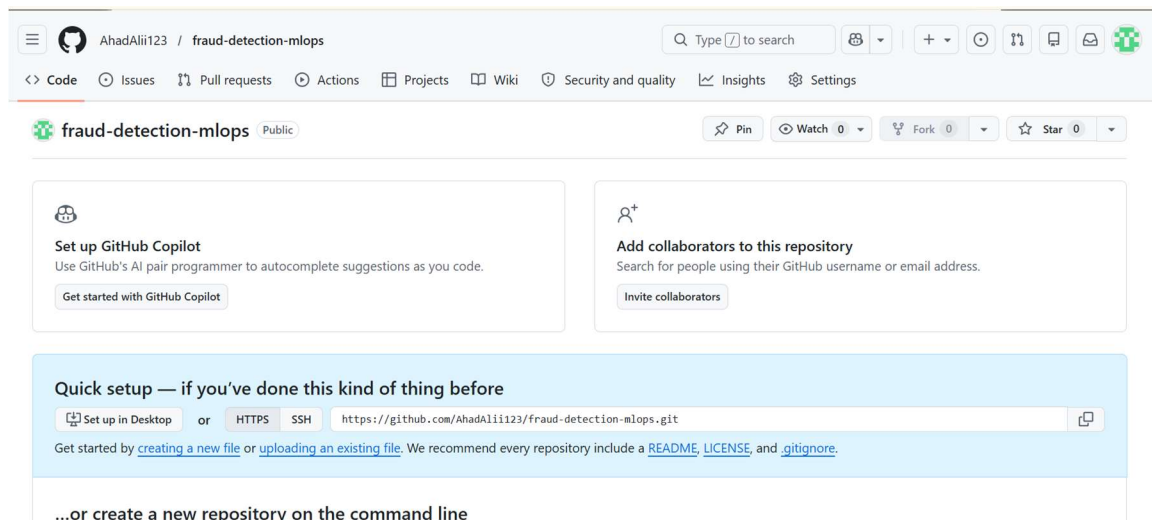
Version Control & Cloud Code Management (GitHub):

Local Git Repository Initialization & Configuration:

```
Ahad Ali@DESKTOP-334LMBR MINGW64 ~  
$ cd "/c/Users/Ahad Ali/Desktop/fraud-detection-mlops"  
  
Ahad Ali@DESKTOP-334LMBR MINGW64 ~/Desktop/fraud-detection-mlops (main)  
$ git init  
Reinitialized existing Git repository in C:/Users/Ahad Ali/Desktop/fraud-detection-mlops/.git/  
  
Ahad Ali@DESKTOP-334LMBR MINGW64 ~/Desktop/fraud-detection-mlops (main)  
$ git add .  
  
Ahad Ali@DESKTOP-334LMBR MINGW64 ~/Desktop/fraud-detection-mlops (main)  
$ git commit -m "Initial commit: End-to-End MLOps Fraud Detection Pipeline"  
On branch main  
nothing to commit, working tree clean
```

Using the Git Bash terminal, the workspace environment was configured to track all core execution scripts, configuration files, and tracking outputs.

Remote Cloud Mapping & Secure Deployment Setup:



A dedicated public repository was created on GitHub to host the project infrastructure.

Quick setup — if you've done this kind of thing before

Set up in Desktop

 or

HTTPS

SSH

https://github.com/AhadAlii123/fraud-detection-mlops.git

Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).

...or create a new repository on the command line

```
echo "# fraud-detection-mlops" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/AhadAlii123/fraud-detection-mlops.git
git push -u origin main
```

...or push an existing repository from the command line

```
git remote add origin https://github.com/AhadAlii123/fraud-detection-mlops.git
git branch -M main
git push -u origin main
```

ProTip!

 Use the URI for this name when adding GitHub as a remote

The local setup was securely linked to this remote repository, ensuring a clean structure and setting up the foundation for potential CI/CD automation pipelines.

Automated Code Push Execution:

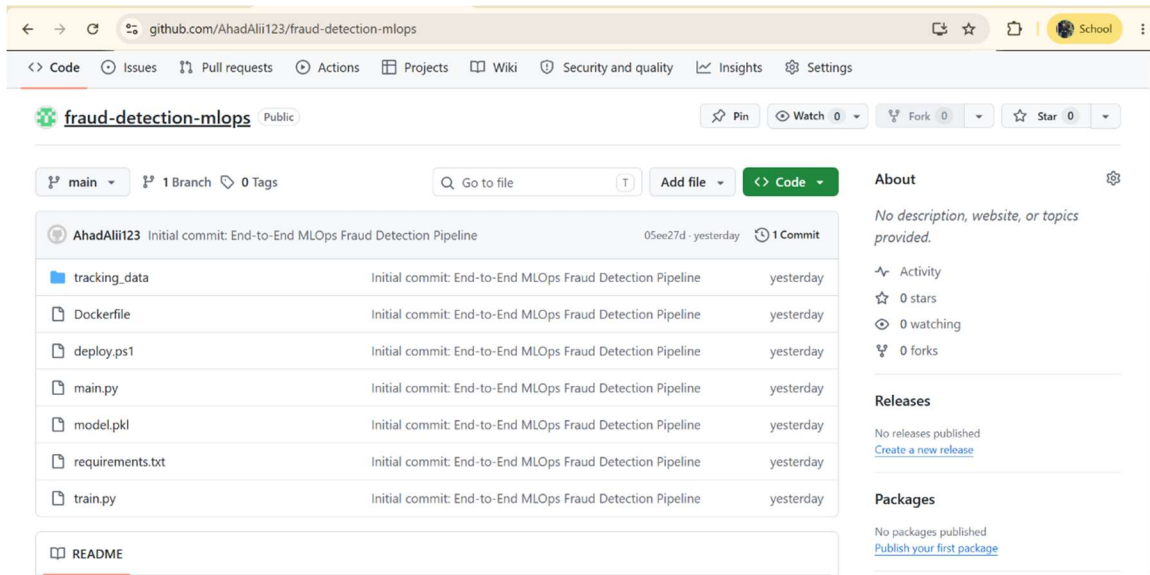
```
Ahad Ali@DESKTOP-334LMBR MINGW64 ~/Desktop/fraud-detection-mlops (main)
• $ git branch -M main

Ahad Ali@DESKTOP-334LMBR MINGW64 ~/Desktop/fraud-detection-mlops (main)
• $ git remote add origin https://github.com/AhadAlii123/fraud-detection-mlops.git

Ahad Ali@DESKTOP-334LMBR MINGW64 ~/Desktop/fraud-detection-mlops (main)
• $ git push -u origin main
Enumerating objects: 46, done.
Counting objects: 100% (46/46), done.
Delta compression using up to 14 threads
Compressing objects: 100% (34/34), done.
Writing objects: 100% (46/46), 105.39 KiB | 3.90 MiB/s, done.
Total 46 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), done.
To https://github.com/AhadAlii123/fraud-detection-mlops.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

The upload completed successfully without any errors or dependency issues.

Verified Live Production Workspace on GitHub:



The screenshot shows the GitHub repository page for 'fraud-detection-mlops' by user 'AhadAlii123'. The repository is public and has 1 branch (main) and 0 tags. The commit history shows an initial commit titled 'End-to-End MLOps Fraud Detection Pipeline' made yesterday. The file list includes 'tracking_data', 'Dockerfile', 'deploy.ps1', 'main.py', 'model.pkl', 'requirements.txt', and 'train.py', all committed yesterday. The right sidebar shows the 'About' section with no description, website, or topics provided, and the 'Releases' and 'Packages' sections with no published items.

github.com/AhadAlii123/fraud-detection-mlops

<> Code Issues Pull requests Actions Projects Wiki Security and quality Insights Settings

fraud-detection-mlops Public

Pin Watch 0 Fork 0 Star 0

main 1 Branch 0 Tags

Go to file Add file <> Code

AhadAlii123 Initial commit: End-to-End MLOps Fraud Detection Pipeline 05ee27d - yesterday 1 Commit

tracking_data	Initial commit: End-to-End MLOps Fraud Detection Pipeline	yesterday
Dockerfile	Initial commit: End-to-End MLOps Fraud Detection Pipeline	yesterday
deploy.ps1	Initial commit: End-to-End MLOps Fraud Detection Pipeline	yesterday
main.py	Initial commit: End-to-End MLOps Fraud Detection Pipeline	yesterday
model.pkl	Initial commit: End-to-End MLOps Fraud Detection Pipeline	yesterday
requirements.txt	Initial commit: End-to-End MLOps Fraud Detection Pipeline	yesterday
train.py	Initial commit: End-to-End MLOps Fraud Detection Pipeline	yesterday

README

About

No description, website, or topics provided.

Activity

0 stars

0 watching

0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

The production code is now live and fully updated on GitHub.

Multi-Container Build Phase:

Pulling Base Images from Docker Hub :

```
Ahad Ali@DESKTOP-334LM8R MINGW64 ~/Desktop/fraud-detection-mlops (main)
$ docker compose up --build -d
time="2026-06-05T19:36:53+05:00" level=warning msg="C:\\Users\\Ahad Ali\\Desktop\\fraud-detection-mlops\\ddocker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 23/23
  ✓ Image grafana/grafana:latest Pulled 76.6s
  ✓ Image prom/prometheus:latest Pulled 30.5s
#1 [internal] load local bake definitions
#1 reading from stdin 635B done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 667B done
#2 DONE 0.0s

#3 [internal] load metadata for docker.io/library/python:3.11-slim
#3 DONE 2.4s

#4 [internal] load .dockerignore
#4 transferring context: 2B done
#4 DONE 0.0s

#5 [internal] load build context
#5 transferring context: 2.00kB 0.1s done
#5 DONE 0.1s

#6 [1/6] FROM docker.io/library/python:3.11-slim@sha256:a3ab0b966bc4e91546a033e22093cb840908979487a9fc0e6e38295747e49ac0
#6 resolve docker.io/library/python:3.11-slim@sha256:a3ab0b966bc4e91546a033e22093cb840908979487a9fc0e6e38295747e49ac0
```

Successful Service Stack Startup:

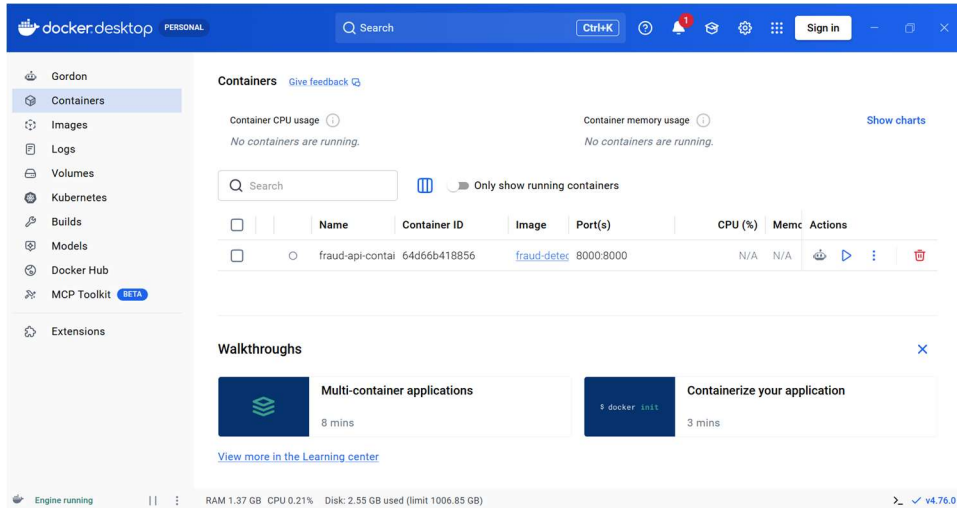
```
Ahad Ali@DESKTOP-334LM8R MINGW64 ~/Desktop/fraud-detection-mlops (main)
$ docker compose up --build -d
#11 CACHED

#12 exporting to image
#12 exporting layers done
#12 exporting manifest sha256:44d6d097da491df285f8d96210b80aad602e3bc3d8428c96e185d6aada67b099 0.0s done
#12 exporting config sha256:9d9153245a45dfb51f7d0afe8185c4d792afb0daccdd146f0903b7c2732019ba2 done
#12 exporting attestation manifest sha256:ddbc644e0d9eb5d40531ab16f594c9ab3fc8bac3aa0052b635fb97cc18d26efa0.0s done
#12 exporting manifest list sha256:2bf65f4897252c1a0a8105a649b3a12e956ca42f13834de68d6fbf94803ac34e
#12 exporting manifest list sha256:2bf65f4897252c1a0a8105a649b3a12e956ca42f13834de68d6fbf94803ac34e 0.0s done
#12 naming to docker.io/library/fraud-detection-mlops-fraud-api-container:latest done
#12 unpacking to docker.io/library/fraud-detection-mlops-fraud-api-container:latest 0.0s done
#12 DONE 0.2s

#13 resolving provenance for metadata file
#13 DONE 0.0s
[+] up 5/5
  ✓ Image fraud-detection-mlops-fraud-api-container Built 2.1s
  ✓ Network fraud-detection-mlops_monitoring_network Created 0.0s
  ✓ Container fraud-api-container Started 1.0s
  ✓ Container prometheus_container Started 1.2s
  ✓ Container grafana_container Started 1.5s

What's next:
  Filter, search, and stream logs from all your Compose services
  in one place with Docker Desktop's Logs view. docker-desktop://dashboard/logs?appId=fraud-detection-mlops
```

Docker Desktop Engine Initialization :



Verification of the Docker Desktop interface showing the global engine state as "Engine Running" before launching the stack.

FastAPI Live System Metrics Stream (Part A) :

```
localhost:8000/metrics

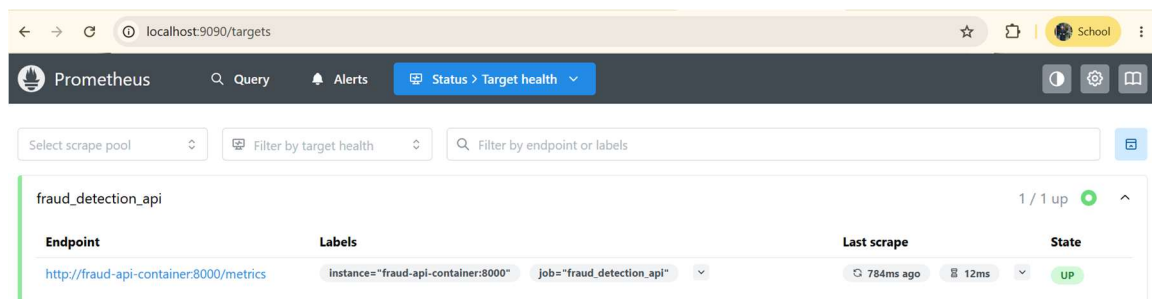
# HELP python_gc_objects_collected_total Objects collected during gc
# TYPE python_gc_objects_collected_total counter
python_gc_objects_collected_total{generation="0"} 772.0
python_gc_objects_collected_total{generation="1"} 458.0
python_gc_objects_collected_total{generation="2"} 20.0
# HELP python_gc_objects_uncollectable_total Uncollectable objects found during GC
# TYPE python_gc_objects_uncollectable_total counter
python_gc_objects_uncollectable_total{generation="0"} 0.0
python_gc_objects_uncollectable_total{generation="1"} 0.0
python_gc_objects_uncollectable_total{generation="2"} 0.0
# HELP python_gc_collections_total Number of times this generation was collected
# TYPE python_gc_collections_total counter
python_gc_collections_total{generation="0"} 370.0
python_gc_collections_total{generation="1"} 33.0
python_gc_collections_total{generation="2"} 3.0
# HELP python_info Python platform information
# TYPE python_info gauge
python_info{implementation="CPython",major="3",minor="11",patchlevel="15",version="3.11.15"} 1.0
# HELP process_virtual_memory_bytes Virtual memory size in bytes.
# TYPE process_virtual_memory_bytes gauge
process_virtual_memory_bytes 1.5322112e+09
# HELP process_resident_memory_bytes Resident memory size in bytes.
# TYPE process_resident_memory_bytes gauge
process_resident_memory_bytes 1.77795072e+08
# HELP process_start_time_seconds Start time of the process since unix epoch in seconds.
# TYPE process_start_time_seconds gauge
process_start_time_seconds 1.78067095067e+09
# HELP process_cpu_seconds_total Total user and system CPU time spent in seconds.
# TYPE process_cpu_seconds_total counter
process_cpu_seconds_total 6.03
# HELP process_open_fds Number of open file descriptors.
# TYPE process_open_fds gauge
process_open_fds 11.0
# HELP process_max_fds Maximum number of open file descriptors.
# TYPE process_max_fds gauge
process_max_fds 1.048576e+06
# HELP http_requests_total Total number of requests by method, status and handler.
```

Validation of the instrumentation output on <http://localhost:8000/metrics>. This confirms the raw stream containing virtual memory bytes, process CPU cycles.

FastAPI HTTP Latency (Part B):

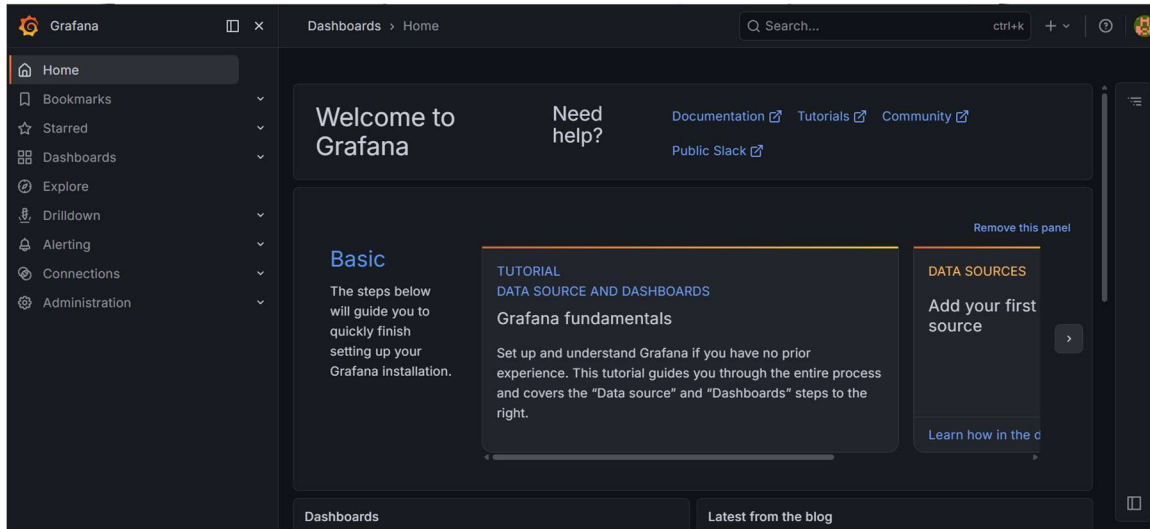
```
http_response_size_bytes_created{handler="/metrics"} 1.7806709607680595e+09
# HELP http_request_duration_highr_seconds Latency with many buckets but no API specific labels. Made for more accurate percentile calculations.
# TYPE http_request_duration_highr_seconds histogram
http_request_duration_highr_seconds_bucket{le="0.01"} 48.0
http_request_duration_highr_seconds_bucket{le="0.025"} 49.0
http_request_duration_highr_seconds_bucket{le="0.05"} 49.0
http_request_duration_highr_seconds_bucket{le="0.075"} 49.0
http_request_duration_highr_seconds_bucket{le="0.1"} 49.0
http_request_duration_highr_seconds_bucket{le="0.25"} 49.0
http_request_duration_highr_seconds_bucket{le="0.5"} 49.0
http_request_duration_highr_seconds_bucket{le="0.75"} 49.0
http_request_duration_highr_seconds_bucket{le="1.0"} 49.0
http_request_duration_highr_seconds_bucket{le="1.5"} 49.0
http_request_duration_highr_seconds_bucket{le="2.0"} 49.0
http_request_duration_highr_seconds_bucket{le="2.5"} 49.0
http_request_duration_highr_seconds_bucket{le="3.0"} 49.0
http_request_duration_highr_seconds_bucket{le="3.5"} 49.0
http_request_duration_highr_seconds_bucket{le="4.0"} 49.0
http_request_duration_highr_seconds_bucket{le="4.5"} 49.0
http_request_duration_highr_seconds_bucket{le="5.0"} 49.0
http_request_duration_highr_seconds_bucket{le="7.5"} 49.0
http_request_duration_highr_seconds_bucket{le="10.0"} 49.0
http_request_duration_highr_seconds_bucket{le="30.0"} 49.0
http_request_duration_highr_seconds_bucket{le="60.0"} 49.0
http_request_duration_highr_seconds_bucket{le="+Inf"} 49.0
http_request_duration_highr_seconds_count 49.0
http_request_duration_highr_seconds_sum 0.21206010699927447
# HELP http_request_duration_highr_seconds_created Latency with many buckets but no API specific labels. Made for more accurate percentile calculations.
# TYPE http_request_duration_highr_seconds_created gauge
http_request_duration_highr_seconds_created 1.7806709552989128e+09
# HELP http_request_duration_seconds Latency with only few buckets by handler. Made to be only used if aggregation by handler is important.
# TYPE http_request_duration_seconds histogram
http_request_duration_seconds_bucket{handler="/metrics",le="0.1",method="GET"} 49.0
http_request_duration_seconds_bucket{handler="/metrics",le="0.5",method="GET"} 49.0
http_request_duration_seconds_bucket{handler="/metrics",le="1.0",method="GET"} 49.0
http_request_duration_seconds_bucket{handler="/metrics",le="+Inf",method="GET"} 49.0
http_request_duration_seconds_count{handler="/metrics",method="GET"} 49.0
http_request_duration_seconds_sum{handler="/metrics",method="GET"} 0.21206010699927447
```

Prometheus Server Target Activation:



The internal target health state dashboard inside Prometheus running at <http://localhost:9090/targets>. It highlights the `fraud_detection_api` scraping endpoint configured correctly under the green **UP** operational state.

Grafana Visualization Management Portal :



The successful deployment confirmation of the analytical UI layer showing the Grafana Central Welcome Portal running at `http://localhost:3000` after securing authentic login bypass.